

CONTROLLED-VOLUME PUMP

Core Working Principle, Graphs and Equations

By Gupta Sahu

Revision 0 | 18 September 2026

Purpose

Explain what a controlled-volume pump is, how suction and discharge strokes create controlled average flow, why the instantaneous flow is pulsating, and how to apply the governing equations during preliminary EPC engineering review.

Document basis and limitations

Item	Basis
Primary equipment reference	API Standard 675, Third Edition, November 2012,
Pulsation reference	API Standard 688, Second Edition, October 2023,
Purpose	Learning and preliminary decision support for controlled-volume pump applications.
Limitation	Not design approval, code certification, acoustic-study replacement, relief validation, or authorization to proceed.
Document precedence	Project PO, MR, specifications, approved datasheets and deviations govern according to the documented project hierarchy.

Contents

- 1. Definition and engineering purpose
- 2. Construction and major components
- 3. Suction and discharge cycle
- 4. Fundamental displacement and capacity equations
- 5. Stroke kinematics and graphs
- 6. Instantaneous flow waveform
- 7. Capacity control and performance terms
- 8. Pressure, force and power
- 9. Pulsation, frequency and piping response
- 10. Pulsation-dampener action
- 11. Suction pressure and cavitation
- 12. Practical EPC application
- 13. Worked example
- 14. Information Required Before Final Decision
- 15. Design Constraint as per API 675

1. Definition and engineering purpose

Exact engineering question

Can the selected pump and connected piping system deliver the specified chemical rate across all credible operating conditions, with the required accuracy and turndown, without cavitation, unintended flow, overpressure or unacceptable pulsation?

1.1 Definition

API 675 defines a controlled-volume pump as a reciprocating pump in which precise volume control is provided by varying the effective stroke length. The same pump family is commonly described as a metering, dosing, proportioning or chemical-injection pump.

The essential idea is simple: the pump traps liquid during a suction stroke and displaces liquid during a discharge stroke. Average flow is controlled by the useful volume per stroke and the number of strokes per unit time.

1.2 Scope distinction

API 675 covered liquid-end arrangements are hydraulic diaphragm and packed plunger. Rotary positive-displacement pumps and direct mechanically actuated diaphragm pumps are outside that specific scope.

Feature	Controlled-volume pump meaning
Motion	Reciprocating
Flow principle	Non-return valves convert alternating chamber flow into one-way delivery
Controlled quantity	Average delivered volume per unit time
Primary controls	Actual/effective stroke length and/or stroke rate
Typical services	Chemical injection, dosing, blending and low-flow high-pressure delivery

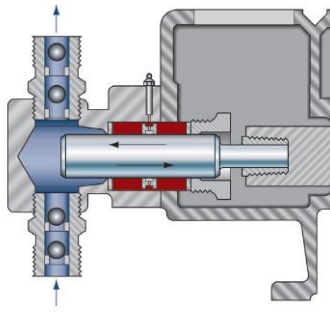
2. Construction and major components

Component	Function	Engineering concern
Driver and gearbox	Create controlled reciprocating motion	Speed, torque, lubrication and area classification
Capacity mechanism	Adjust actual/effective stroke or speed	Minimum reliable setting and actuator resolution
Plunger or piston	Creates geometric displacement	Area, stroke, force and packing
Hydraulic diaphragm	Separates process liquid from drive	Compatibility, fatigue, overtravel and rupture detection
Suction check valve	Admits liquid into chamber	Opening delay, gas, solids and seating
Discharge check valve	Releases liquid to line	Backpressure, closure delay and wear

Component	Function	Engineering concern
Pressure-limiting device	Protects against excessive pressure	Set pressure, capacity, accumulation and return destination

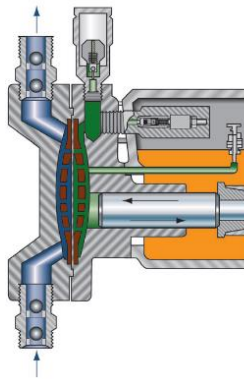
2.1 Two relevant liquid-end concepts

Packed-plunger liquid end: the process fluid directly contacts the plunger and seals against packing.



Packed plunger liquid end

Hydraulic-diaphragm liquid end: a plunger displaces hydraulic fluid, which flexes a diaphragm and isolates the process liquid from the power end.



Hydraulic Diaphragm Liquid End

Practical implication

Diaphragm construction is often preferred where external process leakage is unacceptable. Packed plungers remain attractive for very high pressure when the leakage and maintenance philosophy permits their use.

As per Aramco Standard:

- Up to 35 MPa absolute (5,000 psia) discharge pressure and up to 150°C (302°F): select a diaphragm pump.
- Above 35 MPa absolute (5,000 psia): select either a metal-diaphragm pump or a packed-plunger pump, subject to the exclusions below.
- **Do not select a packed-plunger pump if:**
 - dilution of the process liquid by hydraulic oil is unacceptable;
 - the liquid is extremely hazardous, toxic, or carcinogenic;
 - process leakage to atmosphere is unacceptable; or
 - the process liquid must not contact air.
- Under any of these four conditions, the standard requires a **double-diaphragm pump**.

3. Suction and discharge cycle

3.1 Suction stroke

- The plunger or diaphragm moves away from the chamber.
- Chamber volume increases and chamber pressure falls.
- Suction-line pressure opens the suction check valve.
- Liquid enters while the discharge check valve remains closed.

3.2 Discharge stroke

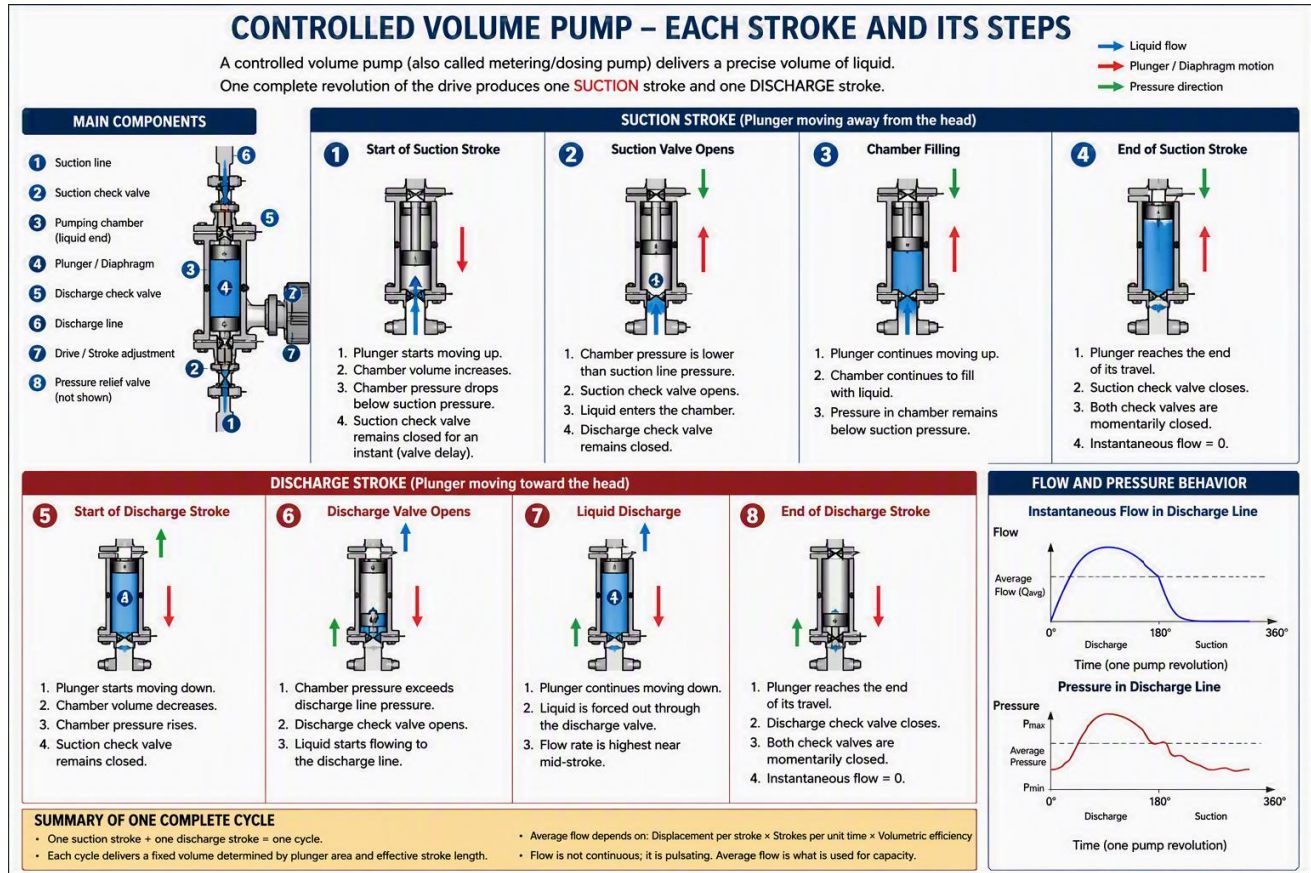
- The plunger or diaphragm moves toward the chamber.
- Chamber volume decreases and pressure rises.
- The suction check valve closes.
- When chamber pressure exceeds discharge pressure, the discharge check valve opens and liquid is expelled.

3.3 Stroke reversal

At each dead centre, plunger velocity becomes zero momentarily. Real valves do not open or close instantaneously. Valve mass, spring force, viscosity, pressure pulsation, clearance volume and trapped gas create delays and modify the ideal flow waveform.

Core physical model

The moving element creates displacement. The check valves determine direction. The suction system determines whether the chamber fills. The discharge system determines the pressure required to expel the trapped volume.



4. Fundamental displacement and capacity equations

4.1 Plunger area and displacement

For a circular plunger:

$$A_p = \frac{\pi d_p^2}{4}$$

where:

- A_p = plunger area;
- d_p = effective plunger diameter.

The geometric displacement per full effective stroke is:

$$V_s = A_p L_e$$

where:

- V_s = volume displaced per stroke;
- L_e = effective stroke length.

Therefore:

$$V_s = \frac{\pi d_p^2}{4} L_e$$

4.2 Theoretical and actual capacity

If the pump makes N strokes per minute, theoretical average flow is:

$$Q_{th} = V_s N$$

For z equal pumping heads:

$$Q_{th} = z V_s N$$

Actual delivered flow is represented by:

$$Q_{actual} = z V_s N \eta_v$$

where:

- Q_{actual} = actual average delivered flow;
- η_v = volumetric efficiency;
- z = number of pumping heads;
- N = strokes per unit time.

API Standard 675 defines volumetric efficiency as the ratio of flow at the rated point to total theoretical displacement per unit time.

Thus:

$$\eta_v = \frac{Q_{actual}}{Q_{th}}$$

4.3 Mass-flow conversion

The mass flow rate is calculated as:

$$\dot{m} = \rho Q$$

where:

- $\dot{m}$ = mass flow rate
- ρ = fluid density
- Q = volumetric flow rate

5. Stroke kinematics and graphs

5.1 Slider-crank equations

For an idealized crank mechanism with an infinitely long connecting rod, let:

- r = crank radius
- l = connecting-rod length
- ω = constant crank angular velocity
- θ = crank angle measured from the reference dead-center position
- x = plunger displacement from the reference dead-center position
 - v_p = plunger velocity
 - a_p = plunger acceleration

The plunger displacement, velocity, and acceleration are:

$$x = r(1 - \cos \theta)$$

$$v_p = r\omega \sin \theta$$

$$a_p = r\omega^2 \cos \theta$$

These expressions represent simple harmonic motion and assume that the connecting-rod length is sufficiently large relative to the crank radius.

Correction for a Finite Connecting-Rod Length

For a finite connecting-rod length, the plunger motion is not perfectly sinusoidal. Using the first-order connecting-rod correction, the displacement may be approximated as:

$$x \approx r(1 - \cos \theta) + \frac{r^2}{4l}(1 - \cos 2\theta)$$

The additional term,

$$\frac{r^2}{4l}(1 - \cos 2\theta),$$

is the **second-harmonic correction** associated with the finite ratio of crank radius to connecting-rod length. This correction introduces a slight asymmetry between the two halves of the plunger stroke.

If velocity and acceleration are also required from this approximation:

$$v_p \approx r\omega \sin \theta + \frac{r^2\omega}{2l} \sin 2\theta$$

$$a_p \approx r\omega^2 \cos \theta + \frac{r^2\omega^2}{l} \cos 2\theta$$

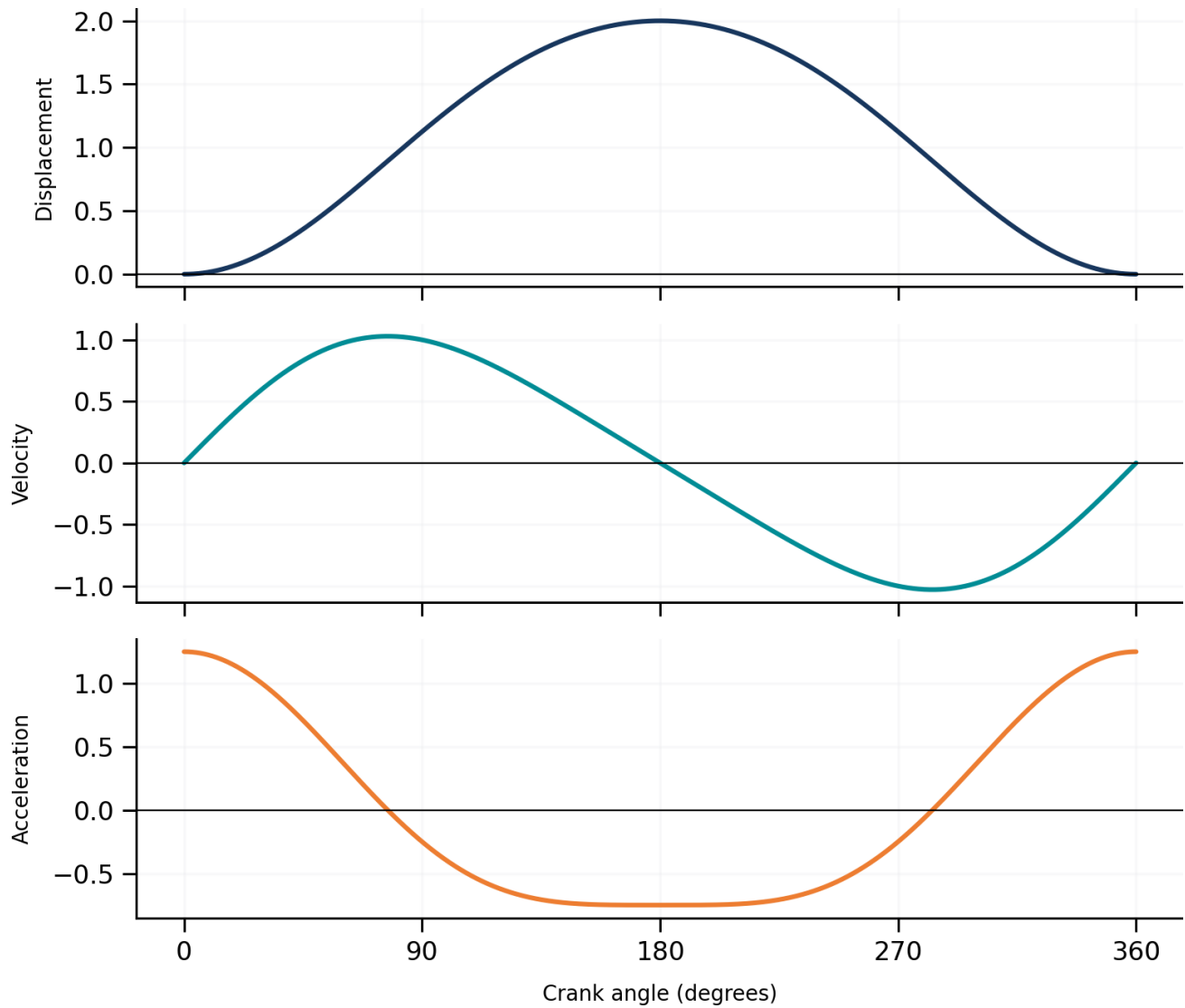


Figure 1. Idealized slider-crank kinematics. Velocity is zero at dead centres; acceleration is highest near reversal.

To understand the above graph below is the complete working example.

The calculation is normalized using:

Parameter	Assumed value	Meaning
Crank radius, r	1	Normalized crank radius
Connecting-rod ratio, r/l	0.25	Therefore $l/r = 4$
Angular velocity, ω	1	Normalized constant crank speed
Full stroke, L	$2r = 2$	Matches the maximum displacement in the graph
Crank-angle reference	0° at first dead centre	Displacement starts at zero

Let:

$$\lambda = \frac{r}{l}$$

For the displayed graph:

$$\lambda = 0.25$$

2.1 Displacement

The approximate finite connecting-rod displacement equation is:

$$x = r(1 - \cos \theta) + \frac{r^2}{4l}(1 - \cos 2\theta)$$

Using $\lambda = r/l$:

$$x = r \left[1 - \cos \theta + \frac{\lambda}{4}(1 - \cos 2\theta) \right]$$

For the normalized graph, $r = 1$ and $\lambda = 0.25$:

$$x = 1 - \cos \theta + 0.0625(1 - \cos 2\theta)$$

2.2 Velocity

Velocity is the time derivative of displacement:

$$v = \frac{dx}{dt}$$

Since:

$$\theta = \omega t$$

the approximate velocity is:

$$v = r\omega \left[\sin \theta + \frac{r}{2l} \sin 2\theta \right]$$

Using $\lambda = r/l$:

$$v = r\omega \left[\sin \theta + \frac{\lambda}{2} \sin 2\theta \right]$$

For $r = 1$, $\omega = 1$, and $\lambda = 0.25$:

$$v = \sin \theta + 0.125 \sin 2\theta$$

2.3 Acceleration

Acceleration is the time derivative of velocity:

$$a = \frac{dv}{dt}$$

The approximate acceleration equation is:

$$a = r\omega^2 \left[\cos \theta + \frac{r}{l} \cos 2\theta \right]$$

Using $\lambda = r/l$:

$$a = r\omega^2 [\cos \theta + \lambda \cos 2\theta]$$

For $r = 1$, $\omega = 1$, and $\lambda = 0.25$:

$$a = \cos \theta + 0.25 \cos 2\theta$$

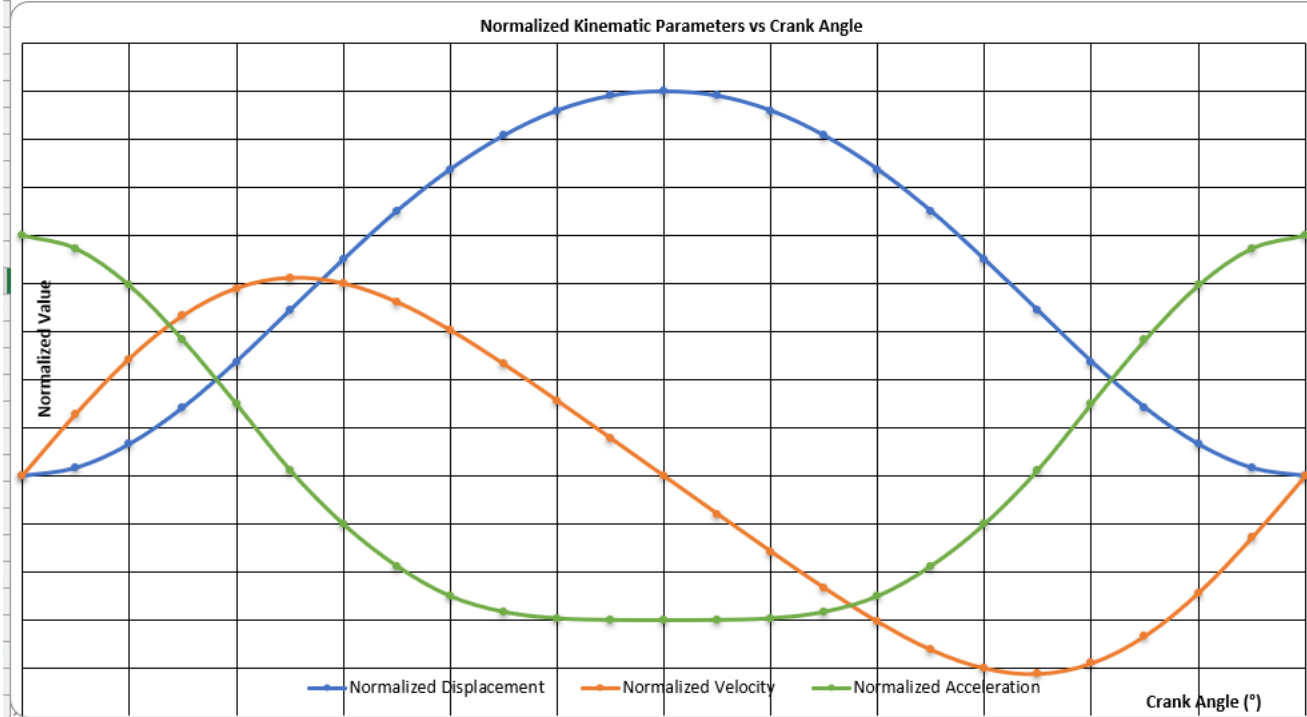
The numerical values below are normalized. Crank angle must be converted to radians when used in spreadsheet or programming trigonometric functions:

$$\theta_{\text{rad}} = \theta_{\text{deg}} \frac{\pi}{180}$$

Crank angle	Normalized displacement x/r	Normalized velocity $v/(r\omega)$	Normalized acceleration $a/(r\omega^2)$	Kinematic interpretation
0°	0.000	0.000	+1.250	First dead centre; plunger begins forward stroke
15°	0.042	+0.321	+1.182	Plunger moving forward and speeding up
30°	0.165	+0.608	+0.991	Forward velocity increasing rapidly
45°	0.355	+0.832	+0.707	Forward motion continues to accelerate
60°	0.594	+0.974	+0.375	Near maximum forward velocity
75°	0.860	+1.029	+0.032	Almost maximum forward velocity

Crank angle	Normalized displacement x/r	Normalized velocity $v/(r\omega)$	Normalized acceleration $a/(r\omega^2)$	Kinematic interpretation
90°	1.125	+1.000	-0.250	Plunger still moving forward but now slowing
105°	1.377	+0.904	-0.468	Forward motion decelerating
120°	1.594	+0.758	-0.625	Approaching opposite dead centre
135°	1.770	+0.582	-0.707	Forward velocity decreasing
150°	1.900	+0.392	-0.741	Strong deceleration near stroke end
165°	1.977	+0.197	-0.749	Near opposite dead centre
180°	2.000	0.000	-0.750	Maximum displacement; plunger reverses
195°	1.977	-0.197	-0.749	Plunger returning and speeding up
210°	1.900	-0.392	-0.741	Return speed increasing
225°	1.770	-0.582	-0.707	Return motion continues to accelerate
240°	1.594	-0.758	-0.625	Plunger returning rapidly
255°	1.377	-0.904	-0.468	Approaching maximum return speed
270°	1.125	-1.000	-0.250	Near maximum negative velocity
285°	0.860	-1.029	+0.032	Return velocity near maximum magnitude
300°	0.594	-0.974	+0.375	Plunger returning but slowing down
315°	0.355	-0.832	+0.707	Return stroke decelerating
330°	0.165	-0.608	+0.991	Approaching first dead centre
345°	0.042	-0.321	+1.182	Strong deceleration near stroke end
360°	0.000	0.000	+1.250	First dead centre; next cycle begins

Normalized Plunger Kinematics versus Crank Angle



The most useful interpretation rule is:

Slope of displacement curve = velocity

and:

Slope of velocity curve = acceleration

5.2 Graph interpretation

Graph	Physical meaning	Project implication
Displacement	Cumulative chamber-volume change	Determines swept volume per stroke
Velocity	Rate of chamber-volume change	Directly drives instantaneous ideal flow
Acceleration	Rate of velocity change	Drives liquid-column inertial pressure and dynamic load

6. Instantaneous flow waveform

6.1 Ideal discharge flow

$$q_d(\theta) \approx A_p r \omega \sin \theta \text{ during the discharge stroke}$$

$$q_d = 0 \text{ during the suction stroke}$$

For a simplex single-acting reciprocating pump, discharge occurs during only one half of the crank cycle. Although the average flow rate remains constant and can be accurately determined, the instantaneous discharge flow rate is highly pulsating. The flow rate starts at zero at the beginning of the discharge stroke, increases to a maximum value near mid-stroke, and then decreases back to zero at the end of the stroke. Consequently, the discharge flow exhibits significant cyclic fluctuations throughout operation.

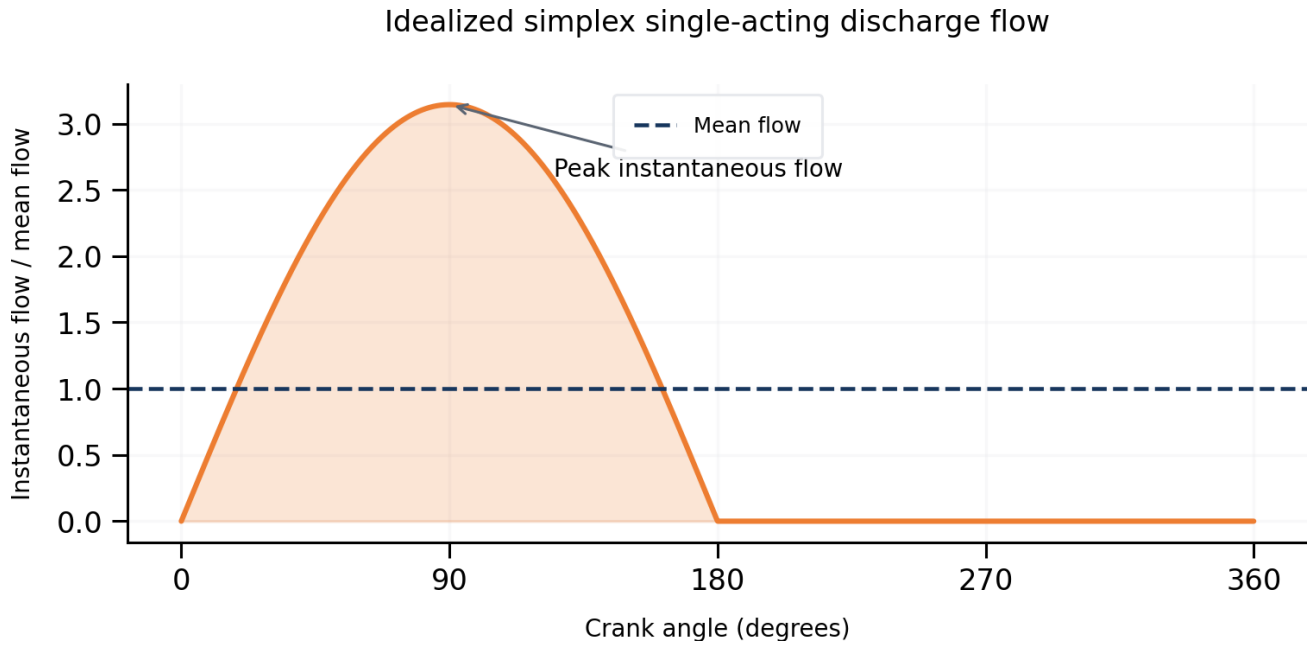


Figure 2. Idealized simplex discharge waveform normalized to its mean. The peak is much higher than average flow.

6.2 Peak-to-average result for the ideal half sine

$$\frac{q_{\max}}{Q_{\text{mean}}} = \pi \approx 3.14$$

This relationship represents an idealized theoretical model of a simplex single-acting reciprocating pump. It indicates that the peak instantaneous discharge flow rate is approximately 3.14 times the mean discharge flow rate. In practice, however, the actual flow characteristics deviate from this ideal behavior due to several influencing factors, including valve opening and closing dynamics, connecting rod geometry, hydraulic compliance of the system, fluid compressibility, pump slip, entrained or trapped gas, and the dynamic response of the connected suction and discharge piping network. Consequently, measured flow pulsations and peak flow rates may differ from the theoretical predictions.

6.3 Multiple heads and phasing

Composite flow improves as phased heads are added

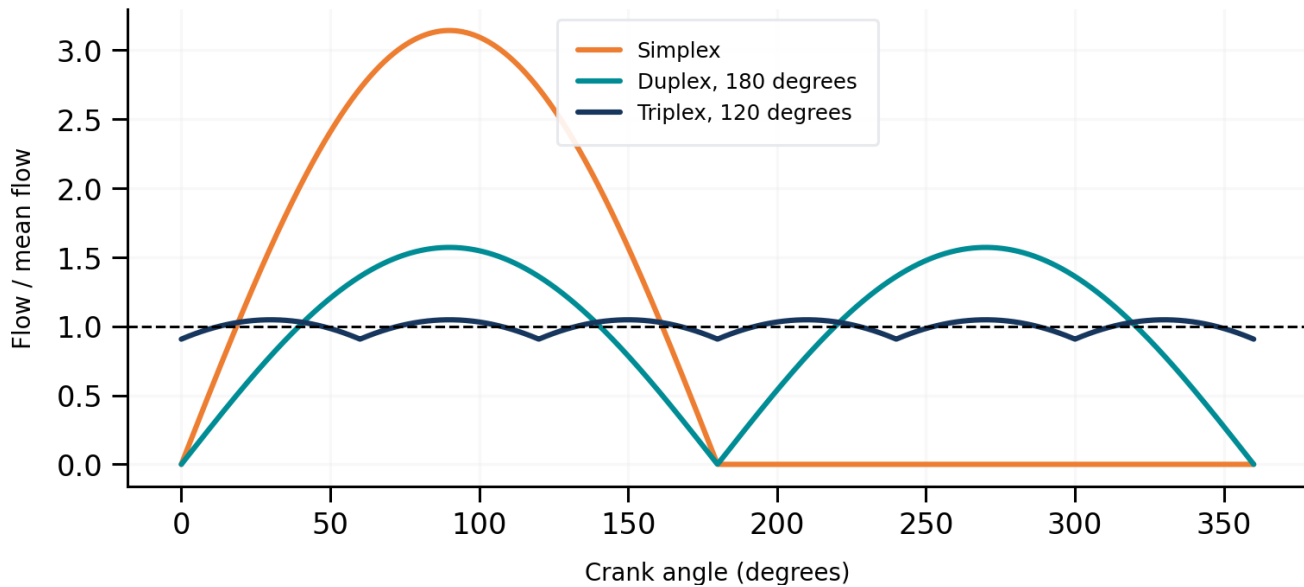


Figure 3. Properly phased multiple heads reduce composite flow variation and raise the pulse repetition frequency.

7. Capacity control and performance terms

7.1 Control relationship

For a reciprocating metering pump, the flow rate is approximately proportional to the effective stroke length, stroking speed, and volumetric efficiency:

$$Q \propto L_e N \eta_v$$

where:

- **Q** = pump capacity (flow rate)
- **L_e** = effective stroke length
 - **N** = stroking speed
 - **η_v** = volumetric efficiency

For two operating conditions, the capacity ratio can be expressed as:

$$\frac{Q_2}{Q_1} \approx \left(\frac{L_{e2}}{L_{e1}} \right) \left(\frac{N_2}{N_1} \right) \left(\frac{\eta_{v2}}{\eta_{v1}} \right)$$

Capacity Adjustment Methods

In accordance with API 675, pump capacity may be adjusted by varying:

- Actual stroke length
- Effective stroke length
- Stroking speed

However, the stroke-setting percentage is not necessarily equivalent to the resulting flow-rate percentage. Changes in volumetric efficiency, hydraulic characteristics, and mechanical tolerances can cause deviations from an ideal linear relationship. Therefore, vendor calibration data and performance testing are essential to accurately establish the correspondence between pump setting and delivered capacity.

Term	Meaning
Rated capacity	Actual quantity delivered at stated operating conditions
Steady-state accuracy	Flow variation under fixed system conditions
Repeatability	Ability to return to the same flow after changing and restoring the setting
Linearity	Closeness of calibration points to a best-fit straight line
Turndown ratio	Rated capacity divided by minimum capacity meeting stated accuracy and linearity

7.2 Attached-edition performance requirements

- **Rated capacity at least 110% of specified maximum capacity.**
- **Steady-state flow accuracy within +/-1% of rated flow over at least 10:1 turndown.**
- **Flow repeatability within +/-3% of rated flow over the specified turndown.**
- **Deviation from linearity not exceeding +/-3% of rated flow over the specified turndown.**

These requirements apply when the attached API 675 edition is contractually invoked and not modified by higher-precedence project documents.

8. Pressure, force and power

The differential pressure across a reciprocating metering pump is defined as:

$$\Delta P = P_{\text{discharge}} - P_{\text{suction}}$$

where:

- $P_{\text{discharge}}$ = discharge pressure
- P_{suction} = suction pressure

The hydraulic force acting on the plunger is approximately:

$$F_{\text{plunger}} \approx \Delta P A_p$$

where:

- F_{plunger} = plunger force
- A_p = effective plunger area

The theoretical hydraulic power required to transfer the fluid is:

$$P_{\text{hydraulic}} = Q \Delta P$$

where:

- Q = volumetric flow rate
- ΔP = differential pressure across the pump

Considering overall pump and drive efficiency, the required input power is:

$$P_{\text{input}} \approx \frac{Q \Delta P}{\eta_{\text{overall}}}$$

where:

- η_{overall} = overall efficiency of the pump-driver system

Blocked-discharge warning

A positive-displacement pump continues attempting to displace liquid into a closed system. An effective pressure-limiting path is required. Closing the downstream valve is not a safe means of reducing flow to zero.

9. Pulsation, frequency and piping response

Positive-displacement machinery creates dynamic flow pulses superimposed on average flow. The piping converts the mismatch between source flow and downstream demand into pressure waves. Pipe length, diameter, fluid density and bulk modulus, valves, vessels, branches and boundary conditions determine the response.

9.1 Primary excitation frequency

$$f_p = \frac{N P n}{60}$$

where:

- N = pump shaft speed (rpm)
- P = number of plungers (or pistons)
- n = 1 for a single-acting pump and 2 for a double-acting pump

9.2 Governing acoustic relationships

Momentum Equation

$$\frac{\rho}{A} \frac{\partial Q}{\partial t} + \frac{\partial P}{\partial x} + RQ = 0$$

where:

- ρ = fluid density
- A = pipe flow area
- Q = volumetric flow rate
- P = pressure
- x = distance along the piping system
 - t = time
- R = hydraulic resistance coefficient

This equation represents conservation of momentum and accounts for fluid inertia, pressure gradients, and hydraulic losses within the piping system.

Continuity Equation

$$\frac{\partial P}{\partial t} = -\frac{k}{A} \frac{\partial Q}{\partial x}$$

where:

- k = fluid bulk modulus
- A = pipe cross-sectional area

This equation relates pressure fluctuations to changes in flow along the pipe and represents conservation of mass for a compressible fluid.

Relationship Between Bulk Modulus and Wave Speed

$$k = \rho a^2$$

where:

- k = fluid bulk modulus
- ρ = fluid density
- a = acoustic wave speed in the fluid

This relationship links fluid compressibility to the speed at which pressure waves propagate through the hydraulic system.

9.3 Mechanical consequence

Dynamic pressure acting at elbows, reducers, tees and closed ends creates shaking force. If the forcing frequency approaches a mechanical natural frequency, vibration may be amplified and can damage supports, valves, instruments and small-bore connections.

10. Pulsation-dampener action

10.1 Storage-and-return mechanism

- High-flow portion: excess pump flow enters the dampener and compresses its gas volume.
- Low-flow portion: compressed gas expands and returns liquid to the line.
- Result: downstream flow and pressure variation are reduced, although not necessarily eliminated.

Conceptual flow smoothing by a pulsation dampener

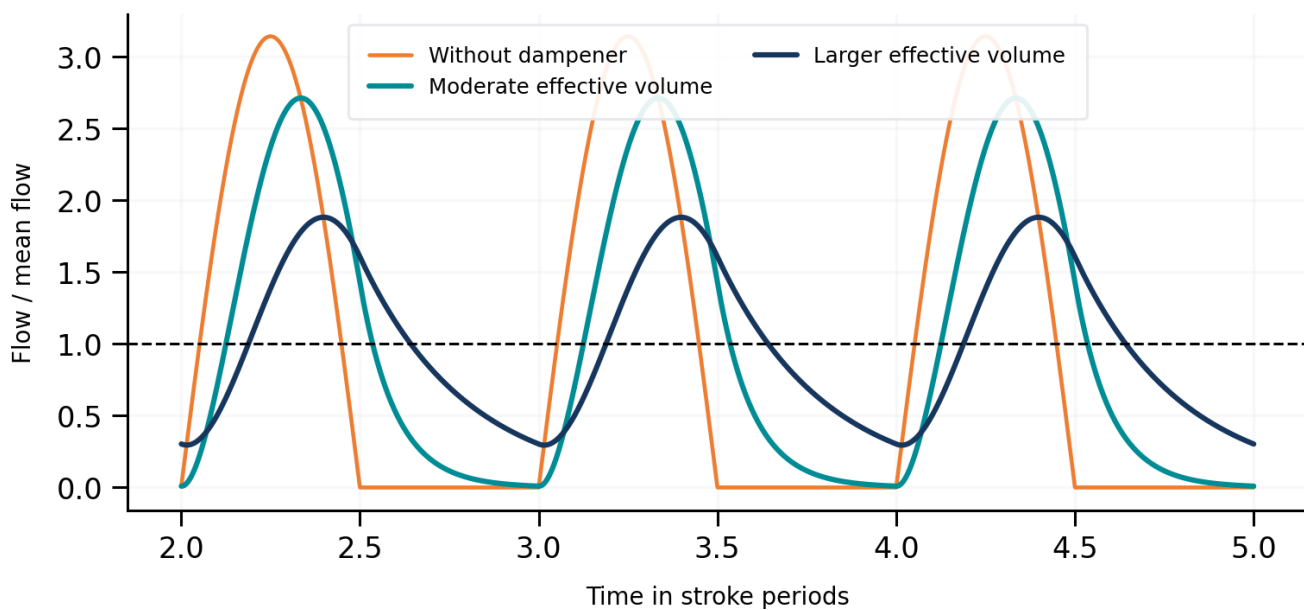


Figure 4. Conceptual dampener response. Greater effective compliant volume produces a smoother downstream waveform.

10.2 Gas-law basis

$$P_1 V_1^m = P_2 V_2^m$$

where:

- P_1, P_2 = initial and final absolute pressures
- V_1, V_2 = initial and final gas volumes
- m = polytropic exponent

Gas-law pressures must be absolute. The exponent m depends on the treatment of heat transfer. Final sizing must use the selected manufacturer method and consider precharge, actual operating pressure range, stroke volume, connection losses, diaphragm or bladder compatibility and residual pulsation target.

10.3 Why a dampener may fail to perform

- Lost or incorrect precharge
- Dampener installed too far from the pump or through a restrictive connection
- Operating pressure far from the design point
- Closed isolation valve or blocked connection
- Incompatible bladder/diaphragm material
- Acoustic resonance in the connected piping
- Sizing based on average hourly flow instead of volume per stroke

11. Suction pressure and cavitation

11.1 Available inlet pressure

For reciprocating metering pumps, the Net Positive Inlet Pressure Available (NPIPA) is a key parameter used to evaluate suction performance and assess the risk of cavitation, vapor release, or loss of pumping accuracy.

$$\text{NPIPA} = P_a + P_z - P_f - P_{vp} - P_{amax}$$

where:

- P_a = absolute pressure at the source liquid surface
- P_z = static suction head (positive) or suction lift (negative)
- P_f = friction losses in the suction piping, fittings, valves, and other components
- P_{vp} = vapor pressure or gas-dissolution pressure of the pumped liquid
- P_{amax} = acceleration-pressure loss resulting from the cyclic acceleration and deceleration of the liquid column in the suction line

$$\text{NPIPA} > \text{NPIPR}$$

Cavitation risk is governed by the negative pressure peak

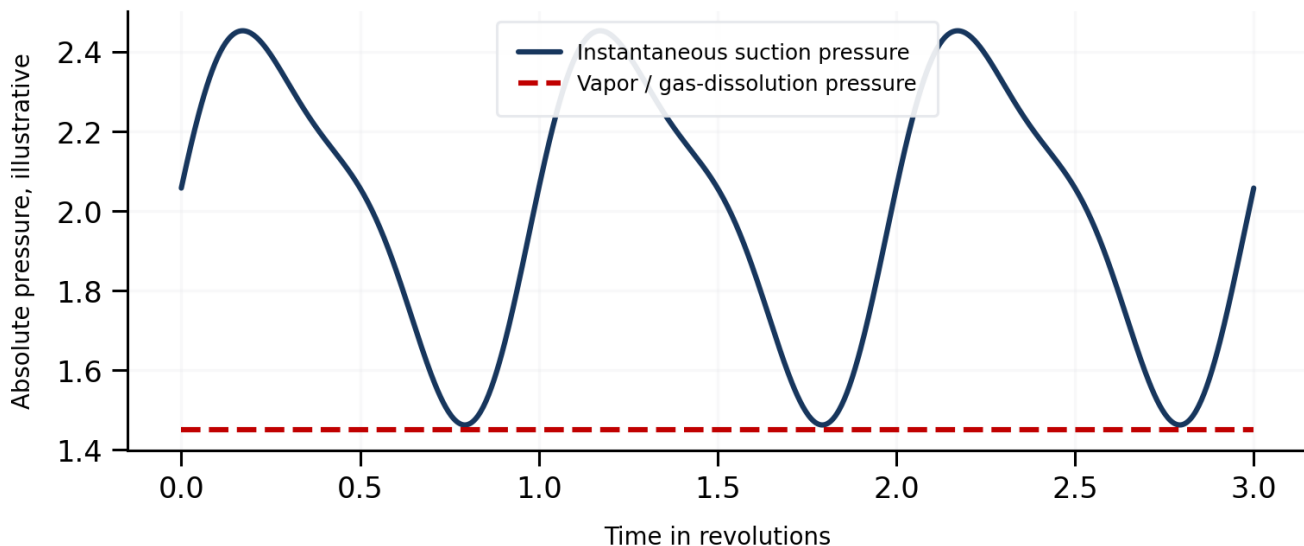


Figure 5. Average pressure alone is insufficient; the negative instantaneous pressure peak controls cavitation risk.

API 688 emphasizes that cavitation can occur when the negative pressure-wave peak approaches vapor pressure, even when a simplified steady inlet calculation appears acceptable. Critical or dynamically complex systems may require acoustic simulation using the actual pump, valve and piping model.

12. Practical EPC application

12.1 What to confirm on the datasheet

- Minimum, normal, maximum and rated flow
- Minimum and maximum suction pressure
- Normal and maximum injection pressure
- Chemical density, viscosity, vapor pressure, concentration, gas and solids
- Effective stroke and maximum stroke rate
- Number of heads and phasing
- Guaranteed turndown, accuracy, repeatability and linearity
- NPIPR and suction calculation basis
- Pressure-limiting, backpressure, antisiphon and pulsation-dampening arrangements

12.2 What to request from the vendor

- Completed API/project datasheet and deviation list
- Capacity versus stroke-setting test curve
- Discharge pressure, NPIPR and power information at specified viscosity
- Displacement per stroke and stroke rate at each duty
- Valve timing or source-flow waveform when a pulsation study is required
- Dampener sizing, precharge and residual pulsation prediction
- Blocked-discharge relief sizing and safe return arrangement
- Performance test procedure and acceptance criteria

12.3 Disciplines to involve

Discipline	Key input
Process	Dose, fluid properties and operating pressure envelope
Mechanical/package	Pump selection, construction, driver and vendor compliance
Piping	Line sizing, layout, supports, nozzle loads and drainability
Instrumentation	Control signal, flow verification, alarms and trips
Process safety	Relief case, return destination and hazardous scenarios
Materials/corrosion	Wetted materials and elastomer compatibility
Operations/maintenance	Priming, calibration, changeover, access and spares

13. Worked example

13.1 Given data

Parameter	Illustrative value
Plunger diameter	20 mm
Effective stroke	12 mm
Stroke rate	100 strokes/min
Number of heads	1
Volumetric efficiency	0.92
Discharge pressure	150 barg
Suction pressure	2 barg
Overall efficiency	0.55

13.2 Capacity Example Calculation: Pump Capacity Determination

Consider a reciprocating metering pump with a plunger diameter of 20 mm, stroke length of 12 mm, stroking speed of 100 strokes/min, and volumetric efficiency of 92%.

1. Plunger Area

$$A_p = \frac{\pi D^2}{4} = \frac{\pi(0.020)^2}{4} = 3.142 \times 10^{-4} \text{ m}^2$$

where:

- A_p = plunger area
- D = plunger diameter = 0.020 m

2. Theoretical Displacement per Stroke

$$\begin{aligned} V_s &= A_p L \\ V_s &= (3.142 \times 10^{-4})(0.012) = 3.77 \times 10^{-6} \text{ m}^3 \\ V_s &= 3.77 \text{ mL/stroke} \end{aligned}$$

where:

- V_s = displacement volume per stroke
- L = stroke length = 12 mm

3. Theoretical Flow Rate

$$\begin{aligned} Q_{th} &= \frac{V_s \times N \times 60}{1000} \\ Q_{th} &= \frac{3.77 \times 100 \times 60}{1000} = 22.6 \text{ L/h} \end{aligned}$$

where:

- Q_{th} = theoretical flow rate
- N = stroking speed (strokes/min)

4. Actual Flow Rate

Accounting for volumetric efficiency:

$$\begin{aligned} Q_{actual} &= Q_{th} \eta_v \\ Q_{actual} &= 22.6 \times 0.92 = 20.8 \text{ L/h} \end{aligned}$$

where:

- η_v = volumetric efficiency = 0.92

13.3 Example Calculation: Plunger Force and Power Requirement

Using the previously calculated actual flow rate of 20.8 L/h and the specified operating pressures:

1. Differential Pressure

$$\begin{aligned}\Delta P &= P_{\text{discharge}} - P_{\text{suction}} \\ \Delta P &= 150 - 2 = 148 \text{ bar} \\ \Delta P &= 14.8 \times 10^6 \text{ Pa}\end{aligned}$$

2. Plunger Load

The hydraulic force acting on the plunger is:

$$\begin{aligned}F &= \Delta P A_p \\ F &= (14.8 \times 10^6)(3.142 \times 10^{-4}) \\ F &= 4.65 \times 10^3 \text{ N} \\ F &\approx 4.65 \text{ kN}\end{aligned}$$

This is the load transmitted through the plunger, crosshead, connecting rod, crank, and gearbox during the discharge stroke.

3. Hydraulic Power

Convert the actual flow rate to SI units:

$$Q = 20.8 \text{ L/h} = 5.78 \times 10^{-6} \text{ m}^3/\text{s}$$

Hydraulic power is:

$$\begin{aligned}P_h &= Q\Delta P \\ P_h &= (5.78 \times 10^{-6})(14.8 \times 10^6) \\ P_h &= 85.5 \text{ W} \\ P_h &= 0.0855 \text{ kW}\end{aligned}$$

4. Input Power

Assuming an overall pump-drive efficiency of:

$$\eta_{\text{overall}} = 0.55$$

the required input power becomes:

$$P_{\text{input}} = \frac{P_h}{\eta_{\text{overall}}}$$

$$P_{\text{input}} = \frac{0.0855}{0.55}$$

$$P_{\text{input}} = 0.155 \text{ kW}$$

14. Information Required Before Final Decision

- Document hierarchy: PO, MR, project specification, datasheets, approved clarifications and deviations
- Exact pump type, number of heads, actual/effective stroke and speed range
- Certified pump capacity, power and NPIPR data at the project fluid conditions
- Complete chemical property envelope
- Final suction and discharge pressure cases
- Final P&ID and piping isometrics
- Dampener model, precharge, sizing basis and residual pulsation
- Pressure-limiting valve set pressure, accumulation, capacity, backpressure and destination
- Control philosophy, flow verification, alarms, trips and duty/standby logic
- Required acoustic/mechanical analysis scope and test requirements

Final engineering takeaway

A controlled-volume pump is a precision reciprocating displacement device. Its average flow is governed mainly by effective volume per stroke and stroke frequency. Its instantaneous flow is inherently pulsating. Check-valve dynamics, inlet pressure, fluid properties, a pulsation dampener and the connected piping determine whether the theoretical displacement becomes accurate and reliable field injection.

15. Design Constraint as per API 675:

15.1 Rated-capacity margin:

As per the API 675 Third Edition, the rated capacity is required to be at least:

$$Q_{\text{rated}} \geq 1.10 Q_{\text{specified maximum}}$$

Therefore, if the maximum specified flow is:

$$Q_{\text{max}} = 35 \text{ L/h}$$

the minimum rated capacity would be:

$$Q_{\text{rated,min}} = 1.10 \times 35 = 38.5 \text{ L/h}$$

This is a minimum API margin, but the project must avoid unknowingly adding several margins, such as process margin, design margin, future margin, and vendor margin, because excessive oversizing can force the pump to operate near its minimum reliable setting.

15.2 Minimum operating point

The pump must cover:

- Minimum required flow
- Normal flow
- Maximum flow
- Rated flow
- Start-up or intermittent flow, where applicable
- Future flow only when contractually approved

The most important selection condition is:

$$Q_{\text{min,project}} \geq Q_{\text{min,guaranteed performance}}$$

A pump with sufficient maximum capacity can still be unsuitable if its minimum project duty falls below the vendor's guaranteed controllable range.

15.3 Performance constraints

As per API 675 Third Edition, the principal controlled-capacity performance constraints are:

Performance property	API constraint
Steady-state flow accuracy	Within $\pm 1\%$ of rated flow
Repeatability	Within $\pm 3\%$ of rated flow
Linearity	Within $\pm 3\%$ of rated flow
Minimum required turndown	At least 10:1 for the stated accuracy range

These values apply to the conditions and test basis defined by the standard and contract. They must not be presented as universally valid at any fluid viscosity, suction pressure, discharge pressure, stroke setting, or installation arrangement.

15.4 Do not mix the performance terms

Steady-state accuracy

Variation in measured flow while the pump operates at a fixed setting and stable conditions.

Repeatability

Ability to return to the same capacity after the setting is changed and later restored.

Linearity

Closeness of measured capacity points to the best-fit straight calibration line.

Turndown

$$\text{Turndown} = \frac{Q_{\text{rated}}}{Q_{\text{min,meeting specified performance}}}$$

A pump can be repeatable but not accurate, or linear without the calibration line passing through zero.

15.5 Minimum differential pressure

The vendor must identify any minimum differential pressure necessary for the specified accuracy.

This is important when:

- Discharge pressure is low
- Suction pressure is relatively high
- The discharge line runs downhill
- Injection-point pressure varies
- The receiving system is depressurized during start-up
- The pump can siphon while stopped

A backpressure valve or antisiphon device may therefore be a system requirement, even though the pump can mechanically operate at low discharge pressure.

15.6 Pressure-design constraints

The pump must be designed for the complete pressure envelope, not only normal injection pressure.

1. Maximum differential pressure

$$\Delta P_{\text{max}} = P_{\text{discharge,max}} - P_{\text{suction,min}}$$

This case governs:

- Plunger load
- Power-end loading
- Gearbox torque

- Driver power
- Diaphragm hydraulic loading
- Valve loading
- Capacity at pressure
- Mechanical integrity

2. Maximum suction pressure

The maximum suction pressure must also be defined because it can govern:

- Liquid-end mechanical loading
- Diaphragm differential loading
- Hydraulic-system operation
- Check-valve differential
- Flow-through while stopped
- Pump vendor allowable suction-pressure limit

3. Minimum differential pressure

$$\Delta P_{\min} = P_{\text{discharge},\min} - P_{\text{suction},\max}$$

This case governs:

- Check-valve stability
- Capacity accuracy
- Forward flow while stopped
- Siphoning
- Need for backpressure or antisiphon protection

The maximum differential-pressure case and minimum differential-pressure case are different, and both must be reviewed.

15.7 Blocked-discharge protection

A controlled-volume pump is a positive-displacement machine. Closing the discharge does not establish a harmless shut-off operating point. Pressure continues to rise until:

- A pressure-limiting device opens
- The driver stalls
- A mechanical component fails
- Leakage becomes significant
- The system loses containment

Therefore, the design must provide an effective pressure-limiting arrangement.

API design implications

- Hydraulic-diaphragm pumps require hydraulic overpressure protection within the hydraulic system.
- Packed-plunger pumps require effective external pressure protection in the connected system.
- A rupture disc is not an acceptable substitute for the required pressure-limiting valve.
- The protective device must pass the maximum credible pump capacity.
- The relief arrangement must not be defeated by isolation valves.

Relief design check

Confirm:

$$Q_{\text{relief capacity}} \geq Q_{\text{maximum credible pump flow}}$$

at the applicable:

- Set pressure
- Accumulated pressure
- Chemical viscosity
- Relieving temperature
- Return-line backpressure

Return destination

The relieved liquid must be routed to an agreed safe destination, such as:

- Chemical tank
- Closed return system
- Recovery vessel
- Other process-approved destination

Do not automatically return relief flow to the pump suction nozzle. That arrangement can create heat buildup, gas generation, concentration changes, and repeated recirculation if the discharge remains blocked.

15.8 Capacity-control mechanism constraints

The capacity-adjustment mechanism may change:

- Actual stroke length
- Effective stroke length
- Stroke frequency
- A combination of stroke and speed

The design should provide:

- Adjustment over the specified operating range
- Accurate, repeatable indication of setting
- Positive locking to prevent unintended change
- Adjustment while operating when required by the applicable design
- Manual and/or remote actuator as specified
- Suitable local indication
- Correct fail position for an automatic actuator
- Adequate actuator thrust or torque
- Hazardous-area suitability
- Defined input-to-capacity calibration

Practical turndown constraint

Avoid selecting a pump whose normal duty is close to the minimum setting.

For example:

- Rated flow: 100 L/h
- Normal flow: 15 L/h
- Minimum flow: 5 L/h

Even if the vendor claims 10:1 turndown, the 5 L/h case gives:

$$\text{Required turndown} = \frac{100}{5} = 20:1$$

The pump would not satisfy a demonstrated 10:1 range without another operating strategy.

One important distinction

There are actually **two different ratios** in this example.

Process operating range :

The process requires flow variation from 15 L/h normal to 5 L/h minimum:

$$\frac{15}{5} = 3:1$$

Therefore, the **process flow variation between normal and minimum is only 3:1**.

Selected pump capability required

Because the selected pump is rated at 100 L/h but must operate at 5 L/h:

$$\frac{100}{5} = 20:1$$

Therefore, the **selected pump requires 20:1 turndown**.

This illustrates that the problem is not the process's 3:1 variation by itself. The problem is that the selected **100 L/h pump is oversized relative to the actual duty**.

That distinction is often important during an MR, datasheet, or vendor bid review.

What pump rating would suit the 5 L/h condition with 10:1 turndown?

If the lowest required flow is 5 L/h and only 10:1 turndown is demonstrated:

$$Q_{\text{max, selectable}} \leq 5 \times 10 = 50 \text{ L/h}$$

Therefore, a pump rated at **50 L/h or less** could theoretically deliver 5 L/h within a 10:1 demonstrated range.

For example, with a 40 L/h rated pump:

$$\text{Setting at normal flow} = \frac{15}{40} \times 100 = 37.5\%$$

$$\text{Setting at minimum flow} = \frac{5}{40} \times 100 = 12.5\%$$

Both operating points would then be above the nominal 10% lower boundary.

However, pump selection cannot be finalized from these three flow values alone. The selected rated capacity must also be accommodated:

- Maximum process flow
- Design or dosing margin
- Chemical concentration variation
- Future operating cases
- Vendor capacity tolerance
- Relief-valve recycled, if applicable
- Actual suction and discharge conditions

6. Acceptable engineering strategies

If a 100 L/h pump is needed for another operating case, possible solutions include:

A. Select a smaller liquid end or pump frame

This is normally the cleanest solution if the true maximum required flow permits it.

B. Obtain a demonstrated 20:1 or greater turndown

Require the vendor to guarantee performance at 5 L/h, not merely state that the pump can be adjusted to 5%.

The guarantee should define:

- Steady-state accuracy
- Linearity
- Repeatability
- Test liquid
- Suction pressure
- Discharge pressure
- Stroke length
- Stroke frequency

- Minimum stable operating point

C. Use combined stroke-length and speed control

The pump can use stroke adjustment for coarse capacity control and variable-speed operation for additional range. The vendor must confirm the permitted speed range, minimum stroking frequency, lubrication limitations, motor cooling, and performance guarantee.

D. Provide two operating pump sizes

For a very wide duty envelope:

- Smaller pump for low-flow operation
- Larger pump for high-flow operation

This may provide better accuracy but adds equipment, piping, controls, maintenance, and cost.

E. Use batch or intermittent dosing

A pump may operate at a stable higher rate for a calculated period rather than continuously at 5 L/h. This is acceptable only if the process permits intermittent dosing and the downstream system provides adequate mixing or buffering.

F. Recycle excess flow

A controlled recirculation arrangement can keep the pump operating above its stable minimum while returning excess chemical. This requires process review because recycling can cause chemical heating, degradation, gas entrainment, or inaccurate net injection.

Suction-system constraints

API 675 treats inlet pressure as a dynamic reciprocating-pump issue, not merely a steady friction calculation.

The inlet assessment can be represented as:

$$NPIPA = P_a + P_z - P_f - P_{vp} - P_{amax}$$

where:

- P_a = absolute pressure at the liquid surface
 - P_z = static liquid head or lift
- P_f = suction-line, valve, strainer, and fitting loss
- P_{vp} = highest vapor pressure or gas-dissolution pressure
- P_{amax} = maximum acceleration-pressure loss

The required condition is:

$$NPIPA > NPIPR$$

The API Errata 2 specifically corrected the inlet-pressure relationship to address the **highest liquid vapor pressure or gas-dissolution pressure**. This means gas release must be considered in addition to ordinary boiling or vaporization.

Suction-layout constraints

The suction line should generally be:

- Short
- Direct
- Adequately sized

- Continuously arranged to prevent gas pockets
- Free of unnecessary fittings and restrictions
- Designed for maximum viscosity
- Designed for minimum absolute inlet pressure
- Evaluated for acceleration loss
- Provided with practical venting and priming connections
- Supported without excessive nozzle loading

Important design principle

Do not size controlled-volume pump suction piping only from average flow.

The peak instantaneous flow of a simplex pump can be several times the average flow, and the complete liquid column accelerates and decelerates every stroke.

15.9 Testing constraints

The vendor must demonstrate performance under the applicable standard and purchaser requirements.

The API 675 performance test includes capacity measurements at multiple capacity points, typically:

- 100%
- 75%
- 50%
- 25%
- 10%

The sequence includes decreasing and increasing settings to assess calibration, linearity, and repeatability.

15.10 API 675 allowable pressure pulsation

For Analysis Approach 1 or 2, API 675 Annex F states that the **peak-to-peak pulsation level beyond the pulsation-control device** should not exceed:

$$P_{1,\text{allow}} = \frac{3500}{\sqrt{ID f}}$$

where:

- $P_{1,\text{allow}}$ = maximum allowable **peak-to-peak pressure pulsation for one frequency component**, kPa
 - ID = inside diameter of the line pipe, mm
 - f = pulsation frequency, Hz

The corresponding US customary equation is:

$$P_{1,\text{allow}} = \frac{100}{\sqrt{ID f}}$$

where:

- $P_{1,\text{allow}}$ is in psi
- ID is in inches

- f is in Hz

These limits apply in the suction and discharge piping **beyond the pulsation-control device**, not inside the pump liquid chamber or at an arbitrary position upstream of the dampener

Worked example

Assume:

- Discharge pipe inside diameter:

$$ID = 25 \text{ mm}$$

- Pump speed:

$$N = 120 \text{ rpm}$$

- Fundamental harmonic:

$$n = 1$$

Step 1: Determine pulsation frequency

$$f = \frac{120 \times 1}{60}$$

$$\boxed{f = 2 \text{ Hz}}$$

Step 2: Calculate allowable peak-to-peak pulsation

$$P_{1,\text{allow}} = \frac{3500}{\sqrt{25 \times 2}}$$

$$P_{1,\text{allow}} = \frac{3500}{7.071}$$

$$\boxed{P_{1,\text{allow}} \approx 495 \text{ kPa}}$$

Therefore:

$$\boxed{P_{1,\text{allow}} \approx 4.95 \text{ bar peak-to-peak}}$$

If the mean discharge pressure is 100 barg and the waveform is approximately symmetrical:

$$P_{\text{max}} \approx 100 + \frac{4.95}{2}$$

$$P_{\text{max}} \approx 102.48 \text{ barg}$$

$$P_{\text{min}} \approx 100 - \frac{4.95}{2}$$

$$P_{\text{min}} \approx 97.53 \text{ barg}$$

This corresponds approximately to:

$$\boxed{\pm 2.48 \text{ bar about mean pressure}}$$

or:

$$\boxed{4.95\% \text{ peak-to-peak relative to 100 barg}}$$

The API limit itself is **4.95 bar peak-to-peak for that frequency component**, not “4.95%.” The percentage changes with mean operating pressure.

5. Check all relevant harmonics

Using the same 25 mm ID and 120 rpm:

Harmonic	Frequency	API allowable peak-to-peak component
$n = 1$	2 Hz	4.95 bar
$n = 2$	4 Hz	3.50 bar
$n = 3$	6 Hz	2.86 bar
$n = 4$	8 Hz	2.47 bar
$n = 5$	10 Hz	2.21 bar

The allowable magnitude reduces as frequency increases:

$$P_{1,\text{allow}} \propto \frac{1}{\sqrt{f}}$$

This is important because a dampener may strongly reduce the fundamental pulse but a piping resonance may amplify a higher harmonic.

References

1. API Standard 675, Positive Displacement Pumps - Controlled Volume for Petroleum, Chemical, and Gas Industry Services, Third Edition, November 2012; Errata June 2014 and April 2015; reaffirmed July 2021.
2. API Standard 688, Pulsation and Vibration Control for Positive Displacement Machinery Systems for Petroleum, Chemical, and Natural Gas Industry Services, Second Edition, October 2023; Errata 1, March 2024.